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Development of a Climate Risk Index for Insurance Pricing

Research Project (TER)

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Abstract

In the context of increasing climate disruption, the assessment of environmental risks has become a major challenge for the insurance sector. As part of our Research Project, supervised by Anne Eyraud, we chose to construct a climate risk index in order to evaluate its potential for insurance pricing.

Our approach was structured in several stages. We first identified relevant criteria to include in the calculation of the index. We then explored the fundamentals of insurance pricing to adapt our methodology to the specificities of the sector. Finally, we searched for databases containing as many of the selected criteria as possible.

Given the complexity of this task, we opted for a simplified numerical application: we developed a climate risk score applied to the French regions. This score provides a concrete illustration of how environmental indicators can be used in a pricing-oriented actuarial framework.

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Introduction

In 2023, natural disasters cost insurers in France €6.5 billion, according to France Assureurs, making it the third most expensive year for climate-related claims after 1999 and 2022. These figures reflect an alarming trend: over the past 30 years, the average annual cost of climatic events has more than doubled, highlighting the accelerating impacts of climate change. Events that were once exceptional are becoming more frequent, more intense, and consequently, more costly.

Faced with this reality, the insurance sector faces a major challenge: how can insurers continue to provide effective protection to policyholders in the context of increasingly unpredictable and severe climate risks?

As actuarial students, we are directly concerned by these issues. Our future profession will require us to design pricing models that incorporate the new realities of climate risk. It is therefore essential to rethink traditional tools to ensure their continued relevance, while also developing new approaches that allow for more accurate and appropriate assessment and pricing of climate-related risks.

How can we construct a meaningful climate risk score, and how can it be integrated into an index-based pricing approach to better address the challenges posed by climate change?

To answer this question, our work is structured into three main sections. The first chapter lays the theoretical foundations necessary to understand our topic, by detailing the various types of climate risks and the principles of insurance pricing. The second chapter focuses on building a climate risk score at the regional level, using an approach based on spatially-disaggregated data. Finally, the third chapter outlines a pricing methodology that incorporates this score and presents a concrete application through a case study. This final section illustrates how the score could be integrated into the pure premium calculation of a climate insurance contract.

Chapter 1

Theoretical Framework: Climate Risks and Insurance Pricing

1.1 Climate Risks and Economic Impact

1.1.1 Definition of Climate Risks and Their Evolution

Climate risks refer to the direct and indirect consequences of climate change on the population, the environment, and the standard of living. The risks associated with climate change are likely to increase extreme weather conditions.

Firstly, **natural risks** can be defined as any form of danger or harm caused by physical, chemical, and biological forces intrinsically related to the Earth. These are primarily generated by hydrometeorological phenomena (floods, storms, and frost), geophysical phenomena (avalanches, landslides, rockfalls), volcanic phenomena (lava flows, explosions), and biological phenomena (diseases).

Additionally, there are **climate risks**, which are the direct or indirect consequences of climate change and can significantly impact the population, the environment, and/or the standard of living. These include: heatwaves, wildfires, or droughts, the intensification of certain extreme weather conditions such as floods, changes in river courses, the retreat of glaciers and their impact on hydrology, the lowering of groundwater levels, the widespread displacement of animal and plant species, and an increase in diseases transmitted by harmful organisms. (GARDER CEUX QUI NOUS CONCERNENT LE PLUS).

These climate risks are not without consequences, particularly in sectors such as health, agriculture, fishing, industry, etc.

First of all, they impact the population. Humans are increasingly exposed to the risks of wildfires, floods, heatwaves, and droughts.
The environment also suffers from these climate risks, particularly terrestrial and marine ecosystems. The melting of glaciers alters the hydrological regime of major rivers, while the decrease in ocean levels leads to increased coastal erosion.
These climate-related risks will subsequently impact living standards. According to some studies, the estimated cost of climate disasters is expected to be twenty times higher by 2050. This will have a significant impact on the global economy, particularly in terms of covering the material, human, and natural damages caused by these events. Poverty will be exacerbated by the loss of livelihoods, with a potential combination of destruction

caused by floods, landslides, wildfires, and other associated risks. In any case, there is a high probability that climate change will significantly increase the level of social and economic vulnerability.

1.1.2 Consequences for the Insurance Sector

Global warming has now become a major challenge for the insurance sector, amplifying the uncertainty faced by insurance scheme promoters. Natural disasters lead to a colossal increase in costs for insurers. According to the latest report from *France Assureurs*, climate disruption is now considered the top insurance risk, competing with cyberattacks. The uncertainty surrounding pricing and the increasing frequency of disasters is pushing insurers to raise their indemnities.

In response to these risks, the insurance sector has had to adapt and has implemented proactive measures to address the challenges posed by global warming. This includes enhanced **collaboration with climatology experts** to better understand future extreme events. By integrating climate risk modeling into their decision-making processes, insurers can better anticipate and manage current market fluctuations.
Insurers also have a role to play in **promoting sustainable practices**. By supporting initiatives aimed at reducing the carbon footprint, particularly by funding renewable energy projects, insurance companies can not only contribute to combating climate change but also secure their own future.

1.2 Actuarial Approaches to Insurance Pricing

General Principles of Insurance Pricing

Introduction to Pricing

The determination of insurance premiums is based on three key components:

- **The technical interest rate**
- **The mortality table**
- **The loadings**

The technical components of pricing, such as the interest rate and mortality table, along with provisioning, are strictly regulated under French law.

Excerpt from the *Code des assurances*

Article A. 132-18 of the *Code des assurances* states :

> “Life insurance premiums charged by companies shall include the remuneration of the company and shall be based on:
> – a technical interest rate (. . .)
> – mortality tables (. . .)”.

In contrast, the regulatory framework allows for flexibility with respect to loadings, which can therefore serve as a source of competitive differentiation among insurers.

Concepts of Net Premium and Loadings

Net Premium

The net premium represents the amount calculated at $t = 0$ (the start of the contract) that enables the insurer to guarantee a given coverage.
The underlying principle is the equality of the expected present values of the respective commitments:

- **EPV(A)**: insurer's liabilities
- **EPV(a)**: policyholder's obligations

The net premium can take the following forms:

- **Single premium**: paid in one lump sum.
- **Periodic premium**: paid at regular intervals.

More broadly, the insurer must ensure that the premium can cover:

- Expected claims (cost of benefits)
- Administrative and operational expenses
- A profit margin

Thus, actuarial calculations are essential to ensure the insurer's solvency while maintaining competitive offerings.

Key Formulas

$$\text{Net Premium} = \text{Frequency} \times \text{Average Cost}$$
$$\text{Gross Premium} = \text{Net Premium} + \text{Expenses} + \text{Profit Margin}$$

Loadings

Although the net premium covers the insurer's commitments toward the policyholder, it does not account for commercial and administrative costs.
These costs are incorporated as loadings to ensure the financial viability of the contract.

Classification of Liabilities

Insurance liabilities may differ according to the nature of the risk:

In the event of death

- Immediate or deferred benefits
- Temporary or lifelong coverage
- Payment schedule (beginning, middle, or end of year)
- Fixed or variable benefit

In the event of survival

- Annual or fractional annuities
- Payable in advance or in arrears
- Immediate or deferred
- Temporary or lifelong coverage
- Fixed or variable benefit

1.2.1 Traditional Risk Segmentation Models

Principle of Homogeneous Risk Grouping (Solvency II)

Risk segmentation must be carried out based on homogeneous groups that share similar characteristics. This approach encompasses:

- Underwriting policies,
- Claims profiles,
- Anticipated policyholder behavior,
- Product features, such as embedded guarantees and future management actions.

The main objective is to ensure reliable valuation of technical provisions while maintaining consistency with the specific risk profile of each group.

Segmentation in Non-Life Insurance (Solvency II)

Non-life insurance obligations are segmented by **lines of business**, including:

- Accident,
- Sickness,
- Motor third-party liability,
- Fire and other damage to property,
- Credit and suretyship,
- Assistance, among others.

This segmentation aims to better reflect the nature of the underlying risks and facilitate tailored risk management according to the specific insurance type.

Segmentation in Life Insurance (Solvency II)

Life insurance contracts are classified into **16 segments** based on several criteria, such as:

- Participation in profits,
- Contracts where investment risk is borne by the policyholder,
- Contracts without profit participation,
- Dominant risks, including death, survival, morbidity, or savings.

This classification enables accurate tracking of the technical nature of the associated risks.

Combination of Life and Non-Life Risks

For contracts that cover both life and non-life risks, or multiple types of risks, obligations must be unbundled to ensure correct attribution by line of business. This approach guarantees that each component is evaluated in accordance with its respective risk profile.

1.2.2 Limitations of Traditional Approaches in the Face of New Climate Risks

1. **Insufficient Historical Models**: Early risk modeling methods based on historical data cannot effectively anticipate natural disasters exacerbated by climate change. New models combine historical data with modern computational techniques for more accurate forecasting.

2. **Changes in Climate Risks**: Events such as droughts, floods, or Mediterranean cyclones (*medicanes*) are becoming more frequent or intense, posing challenges for insurers, particularly in densely populated areas.

3. **Reinsurance and Financial Markets**: Insurers rely on reinsurers and the financial market (such as "catastrophe bonds") to absorb the costs of major disasters, but these systems also present risks, such as volatility or bubble effects.

4. **Public-Private Partnerships**: Some governments, like France, collaborate with insurers to manage uninsurable risks through natural disaster regimes, involving uniform premiums and incentives for prevention.

5. **Insurance Innovation**: Solutions like index-based insurance or climate derivatives address short-term climate anomalies, especially supporting vulnerable populations in developing countries.

1.3 Integration of Climate Dimension in Pricing

1.3.1 Literature Review on the Use of Climate Data in Insurance

With the worsening of climate-related phenomena due to climate change, the insurance industry faces significant challenges in assessing, modeling, and managing the associated risks. This requires a strategic use of climate data to ensure the sustainability and adaptability of the sector.

The Different Categories of Climate Risks

The climate risks insurers face can be grouped into three main categories:

1. **Physical Risks**: These include the direct impacts of extreme weather events, such as storms, floods, droughts, and wildfires. These phenomena directly affect the insured properties and populations.
2. **Transition Risks**: Associated with the shift to a low-carbon economy, these risks lead to potential asset value losses due to restrictive energy policies or a change in consumer preferences for more sustainable solutions.
3. **Liability Risks**: These risks emerge when companies are held responsible for their contribution to climate change, for example through legal actions.

These three categories interact with other financial risks, complicating their management and requiring integrated approaches.

The Importance of Modeling and Evaluation Tools

To address climate challenges, insurers rely on technical innovations and advanced methodologies:

- **Modeling Emerging Risks**: With the evolution of hazards, modeling tools must incorporate new risks, such as coastal erosion or intensified wildfires. These developments enable better forecasting of future claims.
- **Long-Term Climate Scenarios**: Projections up to 2050 or beyond allow for the adjustment of pricing strategies. For example, a gradual increase in premiums (e.g., +0.5% per year for natural catastrophe premiums) is considered a solution to balance rising costs.
- **Use of Granular Data**: Precise information, such as the geolocation of insured assets or the carbon footprint of financial assets, helps refine assessments and optimize risk management.

Strategic and Innovative Approaches

Insurance companies are adopting innovative strategies and solutions to reduce their exposure to climate risks:

- **Insurance-Linked Securities (ILS)**: These alternative financial instruments allow the transfer of certain climate risks to financial markets.

- **Risk Mapping**: By cross-referencing physical, transition, and liability dimensions, insurers can visualize the complex interactions between these risks and better calibrate their strategies.

Governance and Regulation

The role of governance bodies and regulators is crucial in addressing climate challenges:

- **Proactive Regulation**: Authorities integrate the Core Principles of Insurance (CPI) to stabilize the sector while enhancing consumer protection.
- **Climate Stress Tests**: These forward-looking analyses assess the financial impacts of climate scenarios while encouraging better preparedness.
- **International Cooperation**: The exchange of best practices between countries and organizations strengthens insurers' resilience to globalized risks.

Insurers' Commitment to the Green Transition

Insurers are not only observers but also active participants in the ecological transition:

- **Awareness and Prevention**: Insurance companies promote sustainable solutions (green energy, infrastructure resilience) and advise their policyholders on measures to take in the face of local climate risks.
- **Investments in Green Projects**: These institutional investments promote long-term sustainability and climate resilience.
- **Internal Transformation**: Key functions, such as actuarial work and risk management, are gradually incorporating climate challenges, and specialized training is raising awareness among boards of directors on climate strategies.

Economic and Social Impacts

Economic projections emphasize the urgency of action. Without robust climate measures, global GDP could fall by 18% by 2050. To avoid such a catastrophe, premium increases, preventive investments, and public-private collaboration are essential to ensure the long-term viability of the insurance sector.

1.3.2 The Relevance of Statistical and Actuarial Approaches to Refine Risk Segmentation

Risk segmentation is a central process in insurance, allowing premiums and strategies to be tailored to the specific profiles of policyholders while addressing the challenges related to pooling and ethical considerations. Statistical and actuarial approaches play a key role in this process by combining technical precision with enhanced personalization.

Statistical and actuarial approaches provide a robust theoretical basis for improving risk segmentation. Although some of the techniques mentioned here will not be implemented directly in our application, they provide a useful reference framework for future extensions.

Foundations and Principles of Risk Segmentation

Segmentation relies on mathematical and actuarial tools that allow for precise risk classification:

- **Actuarial Principles**: Methods such as *bonus-malus*, credibility, or marginal totals, which adjust premiums according to past claims experience and data specific to each policyholder.
- **Probabilistic Calculations and Actuarial Modeling**: The use of advanced statistical models to estimate financial impacts, necessary reserves, and predictive scenarios.

These techniques ensure appropriate pricing while contributing to overall risk management by better anticipating claims.

Innovations in Segmentation Techniques

Technological advances offer new opportunities to refine segmentation:

1. **GLM (Generalized Linear Models)**: A robust and classical approach that allows clear and understandable segmentation of risks.
2. **Machine Learning (ML)**: Automated and precise techniques that facilitate hyper-segmentation through fine-grained data analysis. Although effective, these tools raise concerns about transparency.
3. **Hybrid Approaches**: A combination of GLM and ML to maximize accuracy while maintaining the robustness and interpretability of results.

Behavioral data, derived from modern technologies such as big data or embedded telematics, also enables advanced algorithmic personalization, steering insurance towards individualized offerings.

Tensions Between Segmentation and Pooling

While segmentation optimizes pricing accuracy, it can weaken the fundamental principle of pooling, which is at the core of the insurance business model:

- **Complementarity Between Segmentation and Pooling**: Segmentation enables fairer pricing, but hyper-segmentation risks fragmenting homogeneous groups, promoting a gradual demutualization.
- **Quantification of Impacts**: Measures such as entropy or variance are used to assess the effectiveness of pooling and understand the balance between fine segmentation and collective solidarity.

Ethical and legal concerns must also be taken into account to ensure transparency, avoid discrimination, and protect policyholders' personal data.

Practical Applications and Perspectives

Statistical and actuarial tools find concrete applications in risk management and pricing:

- **Construction of Technical Premiums**: Developing premiums by considering product data, policyholder profiles, and economic balance.
- **Advanced Tools**: Concepts such as Kolmogorov complexity or information theory enrich the analysis of pooling and understanding of segmentation.

Economic and Strategic Issues

In a changing market context, segmentation must be handled with care to maintain the balance between economic efficiency and collective solidarity:

- **Management of Systemic Risks**: Geographic and sectoral diversification to counterbalance risks associated with concentration.
- **Public Intervention and Reinsurance**: Institutional support to secure high-risk profiles and ensure the insurability of vulnerable populations.

In summary, statistical and actuarial approaches offer powerful tools to refine risk segmentation while addressing the challenges of pooling, personalization, and ethical concerns. A balanced and strategic use of these tools is essential to ensure the sustainability of the insurance sector.

Although complex, these statistical and actuarial tools can be used to devise ways of improving climate pricing. In the context of our work, we have not applied these advanced methods (such as GLM or machine learning), mainly because of data constraints and the exploratory nature of our project. However, the fine segmentation approach, the differentiated weighting logic, and multivariate analysis via PCA are already part of this process of enriching actuarial reasoning. Our approach can therefore be seen as a first step towards the further integration of these tools in future work. Furthermore, a natural extension of our score could be to use these methods to automatically calibrate the weights assigned to variables, or to predict claims on the basis of the regional score.

These foundations and innovations in risk segmentation naturally pave the way for a deeper reflection on the **construction of a climate risk score**, which also mobilizes statistical and actuarial approaches to assess and anticipate the impacts of climate-related phenomena.

Chapter 2

Construction of the Climate Risk Score: A Statistical and Actuarial Approach

2.1 Key Criteria for Developing the Climate Risk Score

To construct a relevant climate risk score, we identified a set of essential criteria that must be present in the dataset. These criteria ensure the score's consistency and incorporate all potentially useful information for an insurer when pricing insurance contracts.

Meteorological and Climatic Criteria

The first set of elements includes **meteorological and climatic criteria**, such as:

- The frequency and intensity of natural disasters (hurricanes, floods, droughts, wildfires, extreme precipitation, etc.).
- Temperature fluctuations throughout the year (heatwaves, late frosts).
- The occurrence and geographical distribution of extreme wind events that could affect a given area.
- Sea-level rise and coastal erosion.

Geographical and Environmental Criteria

The database must also include **geographical and environmental criteria**, such as:

- Exposure to natural hazards (proximity to coastlines, flood-prone zones, fire-prone forests).
- Soil type and permeability (landslide risks, absorption capacity for precipitation).
- Urbanization and vegetation cover (presence of urban heat islands, ecosystem resilience).

Socio-Economic and Infrastructural Criteria

Some less obvious factors have proven to be relevant based on the existing literature. These include **socio-economic and infrastructural criteria**:

- Population density and human exposure to climate-related events.

- Building types and compliance with construction standards.
- Quality of preventive infrastructure (levees, early warning systems).
- Economic vulnerability of residents (resilience capacity after a disaster, ability to evacuate when alerted).

Historical Data and Climate Projections

Finally, although more complex to collect, we aimed to include **historical data and climate projections**, offering a dynamic perspective on risk:

- Trends in claims over time (number and cost of insured losses).
- Observed changes in climate patterns (acceleration of trends, projections from climate models).

2.2 Choice and Analysis of Variables

We aimed to find a dataset that included as many of the previously mentioned variables as possible. However, this search proved more difficult than expected. We therefore chose to build our own database from open and accessible sources.
We limited our scope to French data, which is both more accessible and easier to interpret. Our objective was to aggregate, for each French region, a set of indicators reflecting their level of exposure to climate risk. This work led to the development of a regional climate risk score, usable in the context of differentiated pricing.
Here is the list of variables we retained to build our database and which were used in the calculation of the score:

Frequency of natural disasters: The frequency of natural disasters is a key criterion in the construction of our score. We chose to classify each region on a scale from 1 to 3, where 1 corresponds to the least exposed regions and 3 to the most affected ones. To establish this classification, we relied on a map from the Caisse Centrale de Réassurance (CCR), which uses a color code to illustrate the areas that have experienced the most natural disasters (floods, earthquakes, and droughts). This approach allowed us to assign each region a category based on its level of exposure.
See Figure A.1 in the appendix for the map of the frequency of natural disasters.

Costs of climate risks: To refine our analysis of climate risks, we introduced a new variable: the cumulative cost of losses caused by natural disasters. We relied on the same CCR report as for the previous variable. This data allows us to assess the financial impact of natural disasters in each region. Unlike the frequency of disasters alone, this variable takes into account the economic magnitude of events, thus revealing significant disparities between territories. Some areas, although less affected in number of events—such as Île-de-France—may present very high costs due to urban density or the value of exposed infrastructure.
See Figure A.2 in the appendix for the map of the distribution of climate risk costs.

Average temperature in France: To incorporate average temperature into our analysis, we used data from the website *W3Ask*, which presents a map of average temperatures by department in France between 2018 and 2021. To obtain a representative value at the regional level, we calculated the average of departmental temperatures for each region. See Figure A.3 in the appendix for the map of the average temperature in France.

Climate anomalies: We added a new variable to our analysis: the rise in average temperature, which highlights the various climate anomalies present across the country. This data comes from a map published by the Ministry of Ecological Transition in a brochure dedicated to the impacts of climate change in France. It illustrates the temperature deviation observed in the decade 2000–2009 compared to the reference period 1961–1990.
We chose this variable because it reveals the regions most affected by global warming. Rising temperatures can exacerbate extreme phenomena such as heatwaves, droughts, and water evaporation, thereby directly influencing climate risks and their economic impacts.
See Figure A.4 in the appendix for the map of temperature anomalies in France.

Extreme precipitation: For this variable, we relied on a map from the website *data.gouv.fr*, which shows precipitation accumulations in France and classifies them using a color code indicating deviations from the norm. Extreme precipitation plays a key role in intensifying natural disasters such as floods, landslides, and soil erosion. A region experiencing much higher-than-normal rainfall is more exposed to these risks, which can have a major impact on infrastructure, populations, and the costs of losses.
See Figure A.5 in the appendix for the national precipitation map.

Extreme winds: We focused on strong winds and integrated two key variables to assess their impact across French regions.
The first variable classifies regions according to the number of days with wind gusts exceeding 100 km/h between 1991 and 2020. This data helps identify areas most exposed to extreme winds in the long term, providing an overall view of their frequency.
See Figure A.6 in the appendix for the graph showing the number of days with gusts over 100 km/h per region between 1991 and 2020.
The second variable measures the change in the annual average number of days with gusts over 100 km/h between 1981–2010 and 1991–2020. This approach allows us to understand how the frequency of strong winds is evolving over time and to identify regions where these phenomena are becoming more frequent due to climate change.
See Figure A.7 in the appendix for the graph showing the change in the annual average number of days with gusts over 100 km/h between 1981–2010 and 1991–2020.
To construct these two variables, we used maps from the website *statistiques.developpement-durable.gouv.fr*, which present this data in map format. This information enables us to better understand the risks associated with extreme winds and to refine our regional climate vulnerability index.

Sea level rise: To reflect territorial disparities in the face of rising sea levels, we included sea level rise rates (in mm/year) in our database. This information comes from reliable institutional sources such as the Ministry of Ecological Transition.
The data is available at *geolittoral.developpement-durable.gouv.fr*.

Coastal Erosion : The coastline is a naturally dynamic area; it can either extend or retreat due to various factors (ocean currents, sea level rise, extreme weather events, land development, river flow to the sea, erosion of dune ridges, material extraction, construction of coastal structures that alter sediment exchanges). As a result, the land/sea boundary evolves over time in a highly variable manner with significant territorial disparities. Areas affected by coastal erosion or retreating coastlines are sectors where the sea has advanced inland (receding shoreline position). This is a key variable in developing the score, as the

movement of the coastline endangers buildings and constructions, which are sometimes located in high-risk zones.
See Figure A.8 in the appendix for the national map of shoreline evolution by coastal departments.

Exposure of Populations to Risks : We have chosen to integrate the population exposure index to climate risks into our score because it provides a fairly comprehensive view of the areas where residents are most exposed to different types of hazards (floods, fires, clay shrinkage and swelling, etc.). Figure A.9 in the appendix clearly shows that this exposure varies greatly depending on the territory, particularly affecting densely populated areas. This is an important piece of information to consider in an insurance context, as the more exposed and populated an area is, the more significant the risks for the insurer. Thus, including this variable allows us to add a territorial and human dimension to the climate risk score, making it more relevant for pricing.

Soil Carbon Stock : As a carbon sink, soil helps mitigate climate change. The average organic carbon stock variable is therefore a key indicator to integrate into the development of the climate score, as it reflects the ability of soils to store and sequester atmospheric CO_2. A soil rich in organic carbon plays an essential role in climate regulation by reducing greenhouse gas concentrations in the atmosphere. Moreover, this variable helps assess the vulnerability of territories to extreme climate events: low carbon stocks may indicate greater sensitivity to erosion, desertification, or soil fertility decline. By incorporating this data into our score, we refine the climate risk measurement by considering a determining factor in environmental balance and ecosystem resilience. Unfortunately, we were unable to find relevant information about carbon stocks in overseas territories and Corsica. See Figure A.10 in the appendix for the map of estimated organic carbon stocks in mainland France.

Urban Heat Islands : The inclusion of the urban heat islands (UHI) variable in the climate risk score is based on the significant impact this phenomenon has on local climate conditions, particularly in urban areas. UHIs refer to the elevation of temperatures in cities, which are much higher than in the surrounding rural areas, mainly caused by the presence of artificial surfaces such as concrete and asphalt, lack of vegetation, and human activity. In France, this phenomenon is particularly pronounced in major metropolitan areas such as Paris, Lyon, or Marseille, where temperature differences between city centers and outskirts can reach several degrees, especially during the summer. Exposure to UHIs is therefore a key factor in assessing climate risk, as it affects the quality of life of urban populations, public health, and energy management. Regions with high exposure, such as those containing large urban agglomerations (Paris, Lyon, Marseille, Nice), experience intense UHIs due to their high population density and substantial urbanization. In contrast, some regions with medium-sized cities or rural areas, although experiencing growing urbanization, have moderate exposure to UHIs. Lastly, rural or less urbanized areas, where vegetation cover is more significant, such as Brittany, Normandy, or Limousin, have low exposure to UHIs. This exposure criterion helps refine the segmentation of climate risk and consider the impact of urban phenomena in the development of a more precise risk score.
Unfortunately, we were unable to find an accurate map with concrete data to incorporate this variable into the score calculation. We therefore decided to make approximations with available online data. Regions with the most cities likely to be considered UHI have been marked as having high Urban Heat Island exposure.

Population density : Based on Figure A.11 in the appendix, it appears that highly urbanized areas, particularly in Île-de-France and some overseas departments such as Mayotte and Réunion, exhibit notably high population densities. Conversely, regions such as French Guiana and Corsica show much lower concentrations, reflecting more dispersed settlements and predominantly rural environments. This disparity in population distribution highlights the challenges of territorial planning and urban development, which are essential for anticipating future needs in infrastructure and quality of life.

Air quality : Air quality, expressed in µg/m$^{\mathbf{3}}$, represents the concentration of a specific pollutant (such as PM2.5, NO_2, or O_3) in one cubic meter of air. This unit allows for quantifying air pollution and assessing its impact on human health and the environment. High levels may indicate increased risk for exposed populations.
See Figure A.12 in the appendix for the air quality map.
Based on this visualization, it is evident that urban and industrialized zones tend to show higher concentrations, posing greater risks for local populations. Conversely, rural regions seem less affected, suggesting better living conditions in terms of air quality. This parameter is crucial for anticipating future risks related to climate change, particularly concerning public health.

Building standards (Normes_bat) : For this variable, we considered three different standards, allowing us to assign each region a score based on the number of standards it complies with (we considered their relevance; for instance, the cyclonic standard was not considered for regions with low cyclone risk). These three standards are essential to ensure that buildings are sustainable and adapted to current challenges (source: French government website):

1. **RE2020** : This regulation aims to improve energy efficiency in buildings, reduce carbon emissions, and promote the use of biosourced materials.

2. **Seismic and cyclonic standards** : Seismic standards **ensure** building safety against earthquake risks by requiring specific design and structural criteria. Cyclonic standards are tailored to regions exposed to strong winds and ensure building resistance to cyclones through solid anchoring systems and reinforced structures.

3. **PNACC (National Adaptation Plan to Climate Change)** : This plan provides measures to adapt buildings to the impacts of climate change, such as managing heatwaves, floods, and erosion.

See Figure A.13 in the appendix for the histogram of standards respected.
The chart shows the distribution of regions according to the number of standards complied with, with frequencies concentrated around two standards. This indicates that most regions partially comply with essential standards for ensuring building sustainability and adaptation to climate-related risks. Regions complying with all three standards represent a minority, possibly reflecting better preparedness for environmental risks such as cyclones, earthquakes, or the impacts of global warming. This score can thus serve as a relevant indicator.

GDP per capita (GDP/cap) : Value expressed in euros (€) per individual, representing the average economic output produced by region. It is used to compare levels of economic development. Source: official economic data.
See Figure A.14 in the appendix for the map of GDP per capita.

The map highlights economic disparities across metropolitan and overseas French regions. GDP per capita values, expressed in euros, reveal that regions such as Île-de-France report significantly higher levels, reflecting a dynamic and highly developed economy. On the other hand, more rural or remote regions, particularly some overseas territories, show much lower values, underscoring developmental inequalities.
GDP per capita is a key variable when assessing future challenges related to climate change. Economically strong regions may have more resources to invest in infrastructure and adaptation measures. Conversely, regions with lower GDP may be more vulnerable to climate impacts if they lack sufficient means to anticipate and adapt.

Unemployment rate : Percentage (%) of the labor force that is unemployed but actively seeking work. It indicates the health of the labor market and regional socio-economic challenges. Source: employment statistics (INSEE, local agencies).
See Figure A.15 in the appendix for the unemployment rate map.
The map, color-coded by regional unemployment rates, illustrates socio-economic disparities throughout metropolitan France and the overseas territories. Overseas regions show significantly higher unemployment rates, with darker shades indicating more severe economic challenges, as seen in French Guiana, Mayotte, and Guadeloupe. In metropolitan France, some regions, notably Île-de-France and Grand Est, display lower rates (lighter shades), reflecting more dynamic labor markets.
Hence, regions with high unemployment may be more affected by climate impacts due to increased socio-economic vulnerability. This could limit their ability to invest in resilient infrastructure or adaptation strategies.
Conversely, regions with low unemployment and stronger economies may be better equipped to absorb climate-related costs and adapt to future challenges.

2.3 Construction of the Climate Risk Score

2.3.1 Objective of the Score

The main objective of this score is to enable insurance companies to identify and assess future climate-related challenges within a given territory. Using this indicator, insurers will be able to adapt their contracts and adjust premiums according to the anticipated level of risk. In the field of climate risk, it is widely acknowledged that the analysis must be structured around three complementary and well-defined dimensions:

- **Physical Risk:** Assesses direct exposure to climatic hazards, such as the frequency of disasters, extreme weather events (temperature, precipitation, high winds, sea level rise, etc.), and their immediate impacts.

- **Transition Risk:** Measures a territory's ability to adapt to economic and regulatory changes associated with the transition to a more sustainable economy. This dimension includes indicators such as soil carbon stock, compliance with environmental standards, or GDP per capita.

- **Liability Risk:** Focuses on socio-economic and legal implications, evaluating indicators such as population density and unemployment rate. These elements help insurers anticipate potential liabilities, particularly in the case of major climate-related claims.

Our score ranges from 0 to 1 and is associated with a letter grade from A to D, where 'A' indicates an optimal score and 'D' indicates a poor score.

2.3.2 Implementation of the Score

To implement our scores, we used the Python programming language. We first processed the data by importing our database and creating four *dataframes*, one for each score, in order to retain only the variables relevant for the corresponding score calculation using the `pd.DataFrame()` function. Then, for each score, we followed the steps below:

- **Data normalization** using a multi-criteria method, which is a comparison technique used to determine the best-performing element in a given context. In other words, it allowed us to assign different weights to each criterion: some contribute positively to the score, while others have a negative effect. The resulting value ranges from 0 to 1.

  - If a variable *increases* the score, the formula is:

    $$\text{Score} = \frac{x}{\max(x)}$$

    Example: for the variable `Qualite_air`, a higher value corresponds to a better score:

    ```
    df_Rphysique['Qualite_air'] / df_Rphysique['Qualite_air'].max()
    ```

  - If a variable *decreases* the score, the formula is:

    $$\text{Score} = \frac{\min(x)}{x}$$

    Example: for the variable `Frequence_catastrophe`, a higher value corresponds to a worse score:

    ```
    df_Rphysique['Frequence_catastrophe'].min()
    / df_Rphysique['Frequence_catastrophe']
    ```

- **Weighted average**: to obtain a final score, we applied a weighted average:

  ```
  df_Rphysique.apply(lambda row:  sum(float(row[var]) * poids_Rphysique[var]
  for var in poids_Rphysique) / sum(poids_Rphysique.values()), axis=1)
  ```

- **Scoring with Z-Score**: To define the categories A, B, C, and D, we used the Z-Score to create *bins* around the mean μ and the standard deviation σ. The Z-Score is calculated as:

  $$Z = \frac{x - \mu}{\sigma}$$

  The classes are then defined using the `.qcut()` function in Python, which allows us to divide the values into a specified number of categories. Here, we assigned the following scores:

  - D: low Z-Scores
  - C: average Z-Scores
  - B: high Z-Scores
  - A: very high Z-Scores

  We chose to use 4 categories, and since our database contains 18 regions, the regions are grouped into sets of 4 or 5 for each score.

2.3.3 Results Verification

Once the database was constructed, it was essential to conduct a global verification to ensure that it was coherent and suitable for analysis. To that end, we decided to perform a general analysis of the database.

We first computed a correlation matrix for the variables. The correlation matrix, presented in the appendix (Figure A.16), highlights the linear relationships between the variables in our dataset. Some variables display strong correlations, which may indicate redundancies or logical connections. For example, *Average_ Temperature* is strongly positively correlated with *Climate_ Anomalies* ($\rho = 0.78$) and negatively correlated with *Air_ Quality* ($\rho = -0.79$), which is consistent with the impact of climate change on atmospheric conditions.
We also observe a strong correlation between the *Disaster_ Cost* and their *Frequency* ($\rho = 0.91$), confirming that more frequent disasters generally lead to higher costs. Similarly, *Sea_ Level_ Rise* is positively linked to several other climate risk indicators, such as *Extreme_ Precipitation* ($\rho = 0.79$) and *Coastal_ Erosion* ($\rho = 0.79$), illustrating the complexity and interconnection of the physical phenomena under study.
Conversely, certain variables such as *Unemployment_ Rate* exhibit negative correlations with climate-related variables such as *Average_ Temperature* ($\rho = -0.63$) or *Climate_ Anomalies* ($\rho = -0.53$), potentially reflecting indirect effects of environmental conditions on socio-economic indicators.

We then performed a Principal Component Analysis (PCA) on the full set of variables (Figure A.17 in the appendix) to identify any significant dimensionality reduction opportunities. The cumulative explained variance graph shows that the first four principal components account for approximately 85% of the total variance, a threshold generally considered sufficient for a good representation of the original dataset. From the 10^{th} component onwards, the curve flattens and the marginal gain in explained variance becomes negligible, suggesting redundancy in the initial variables. This PCA thus indicates that a reduced subset of components is sufficient to capture most of the information, which is useful for visualization, classification, or weighting in the construction of a synthetic index.

To get a general sense of how regions are distributed and whether some of them already cluster based on certain criteria, we applied k-means clustering in Python (Figure A.18 in the appendix), using the first two principal components from the PCA. This approach reveals natural groupings of regions based on their climate and environmental characteristics. Four distinct clusters emerge. For example, the overseas regions (Guyane, Mayotte, Réunion, Guadeloupe, and Martinique) form a separate cluster, illustrating a shared specificity in terms of climate risk exposure. Conversely, metropolitan regions such as Île-de-France appear more isolated, possibly reflecting a unique profile across the variables considered. These groupings confirm that some regions share similar risk profiles, supporting the relevance of constructing an aggregated score that takes into account this multidimensional structure. The clustering thus enhances the interpretation of the score, by showing that it can be used not only to assess individual regions but also to identify homogeneous groups in terms of climate vulnerability.

Finally, the code includes visualization steps to analyze the distribution of the scores:

- **Interactive histograms**: These visualizations display the distribution of Z-Scores and their associated labels.

- **Geographical mapping**: Score data was merged with a *GeoJSON* file to map climate risk distribution. The various regions (metropolitan France and overseas territories) are color-coded by label, providing an intuitive reading for insurers.

These visual checks confirmed that our results were coherent with the underlying data.
Having now laid the methodological foundation and explained the score construction in detail, we turn to its practical application and analyze the results obtained, with the goal of assessing its impact on insurance pricing and exploring how it can help tailor contracts to better address climate-related challenges.

Chapter 3

Application and Results: Towards Improved Pricing

In this chapter, we apply the score we previously built to derive actionable results for insurance pricing.

3.1 Analysis of the Score Results

This section presents the results and general interpretations of each of the scores we studied. For each score, we list the variables included, then present the associated map of score results. The maps were produced using Python, based on the method previously detailed.

Physical Risk

The variables used for this score are:
- *Frequence_catastrophe*,
- *Coût_catastrophes*,
- *Température_moyenne*,
- *Anomalies_climatiques*,
- *Precipitations_extremes*,
- *nb_jours_vents_extremes*,
- *evol_nb_jours_vents_extremes*,
- *Zone_exposition_risques*,
- *Ilots_chaleur_urbaine*,
- *Montée_niveau_mer*,
- *Erosion_cotiere*,
- *Qualite_air*

To build a global score from the various physical risk-related variables, we wanted to assign a weight to each variable, since not all are equally important. As we lacked a fully objective method to fix these weights, we tested several approaches.

The first method was an automatic approach based on optimization. The idea was to find the set of weights that makes the final score as "discriminative" as possible, by maximizing its variance between regions: the higher the variance, the more our score differentiates the regions. For this, we used the `minimize` function from the *SciPy* library, with constraints ensuring all weights are positive and sum to 1. This method allowed us to derive a data-driven weighting scheme.

Unfortunately, this optimization approach based on variance maximization proved to be of limited relevance in our context. While it does maximize the score's variability, it does not guarantee that the resulting weights reflect the actual importance of variables in structuring the data. In practice, this method tended to favor variables with high dispersion, regardless of their effectiveness in distinguishing observations.

We therefore opted for a Principal Component Analysis (PCA) on the variables used in the physical risk score. The goal was to statistically extract the most informative latent dimensions. In particular, we relied on the first principal component (PC1), which captures the largest share of variance in the dataset. The loadings associated with each variable in this component reflect their relative contribution to the main structure. To use them as positive weights, we took the absolute values of these coefficients and normalized them so that their sum equals 1.

This method assigns greater weight to variables that play a central role in shaping the data structure, while ensuring the final score is both interpretable and statistically consistent.

This is the method we retained for weighting the variables in the physical risk score calculation, and it is also the one we used for the other scores, as it proved effective.

Finally, here is the map obtained for the physical risk score:

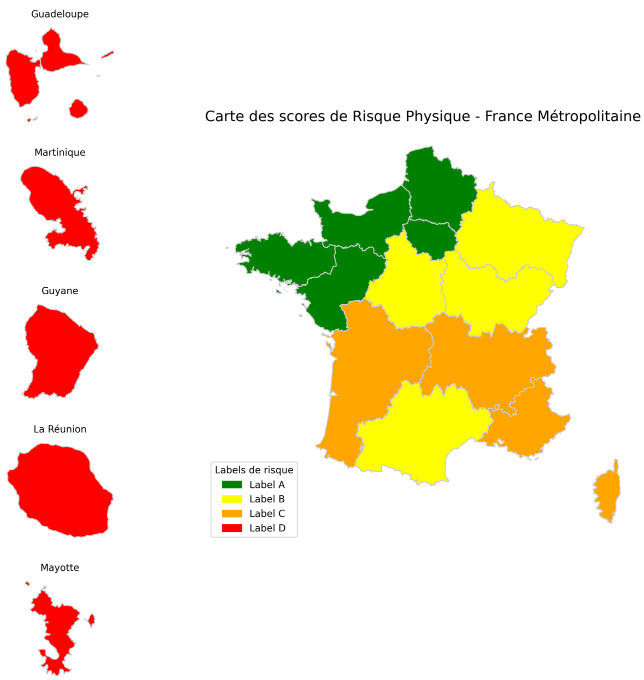

Figure 3.1: Map of the Physical Risk Score

Transition Risk

The variables used for this score are:
- *Stock_carbone_sol*,
- *Normes_bat*,

- *PIB_hab*

Using the same weighting criteria as for the physical risk score, the following results were obtained for the transition risk score:

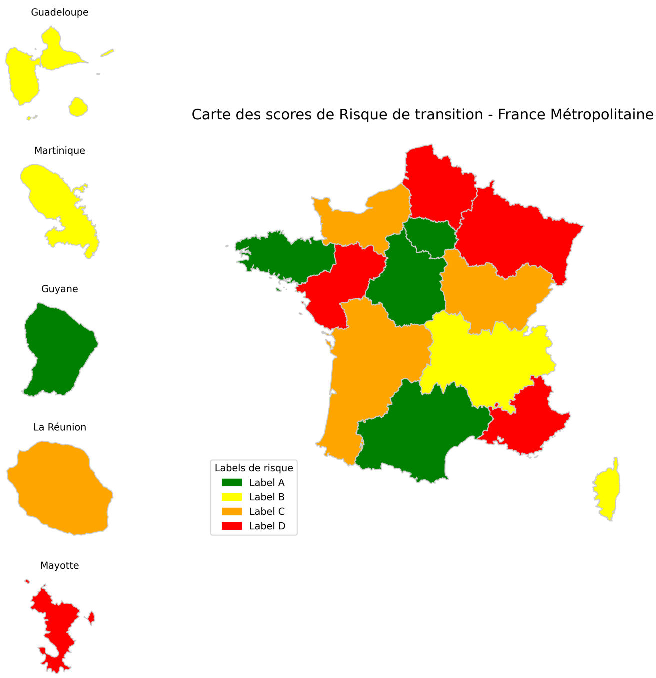

Figure 3.2: Map of the Transition Risk Score

Liability Risk

The variables used for this score are:
- *Densite_population*,
- *Tx_chomage*

Using the same weighting method as for the physical risk score, the following results were obtained for the liability risk score:

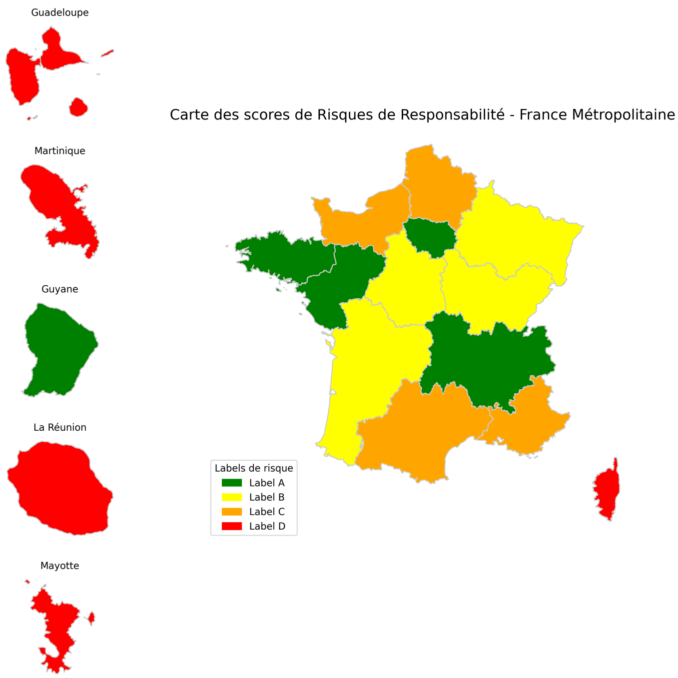

Figure 3.3: Map of the Liability Risk Score

Global Risk

Finally, we decided to conclude with a global risk score, which includes all the variables in our database. The weighting and normalization of each variable were carried out using the same methods as for the other scores.

Here are the results obtained for the global risk score:

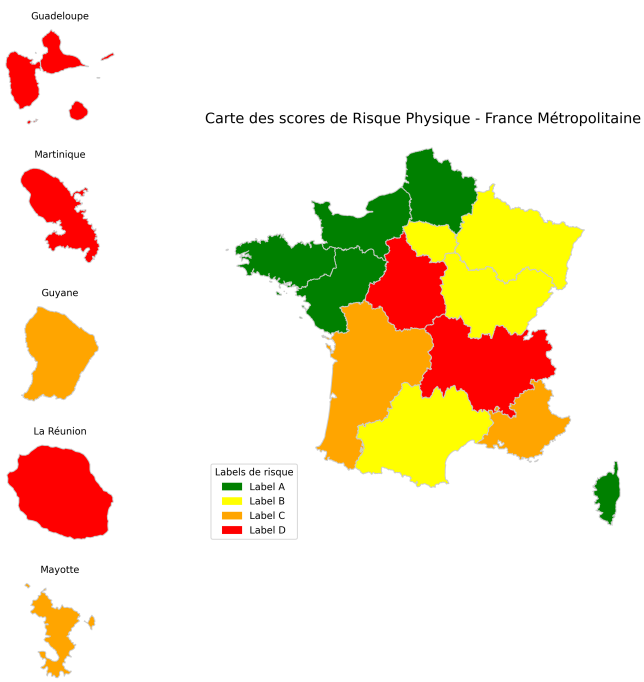

Figure 3.4: Map of the Global Risk Score

3.2 Integration of our index into an insurance scheme

3.2.1 Typology of climate insurance contracts

There are two main insurance mechanisms for covering climate risks:

First, **indemnity-based insurance** is grounded in the actual assessment of damages by an expert. Compensation is proportional to the losses incurred. While this method reflects the real extent of the loss, it involves lengthy processing times and high expertise costs. Moreover, it carries a risk of moral hazard and information asymmetry. Moral hazard refers to a situation in which an individual or company insured against a risk may behave more recklessly than if they were fully exposed to the risk.

Second, there is **index-based insurance**, which relies on a predefined observable index that is independent of actual losses. Compensation is triggered automatically when the index exceeds a predetermined threshold. This method is significantly faster and less expensive. However, it presents a basis risk, which arises when the index is not sufficiently correlated with the actual risk being covered, or when the threshold is poorly calibrated. A poorly chosen index can result in significant costs for the insured.

In our case, the climate index we have designed reflects a **regional exposure level** based on historical data. It does not measure real-time losses. Therefore, it naturally fits into an index-based insurance model.

3.2.2 Stepwise payoff function

To make the best use of our results, we have designed a compensation structure based on a **stepwise payoff function**, meaning that predefined trigger thresholds are set, each corresponding to a specific compensation amount.
Unlike a linear function where every change in the index affects the payout amount, the stepwise function introduces severity levels. This better reflects the actual nature of the risk and ensures a more consistent indemnity scheme.

This approach aligns well with our static and discrete index. It introduces a bonus/malus logic depending on the intensity of the observed risk.

3.3 Pricing method based on our index

3.3.1 Actuarial pricing principle

The pricing of an index-based insurance contract relies on an actuarial method grounded in the historical analysis of the index. The goal is to estimate both the probability that the compensation threshold will be exceeded and the amount to be paid in the event of a trigger.

The pure premium, corresponding to the mathematical expectation of the compensation, is calculated as follows:

$$\text{Pure Premium} = \sum_i \mathbb{P}(\text{Alert}_i) \times (\text{Index} \times \text{Coefficient}_i \times \text{Unit Amount})$$

where $\mathbb{P}(\text{Alert}_i)$ is the probability of occurrence of alert level i,
- the Coefficient_i reflects the severity of the event,
- the unit amount is a fixed monetary amount per index point.

The regional index corresponds to the climate risk score we have established for each region, ranging from 0 to 1, and reflecting their overall exposure to climate hazards.
The unit amount, on the other hand, corresponds to a fixed monetary value assigned to each index point. It allows the conversion of the regional index into a financial estimate of the compensation.

3.3.2 Method application: tiered structure

To adapt our climate index to an index-based insurance scheme, we have defined a tiered compensation function, where the regional score is weighted by a severity coefficient according to the nature and intensity of the observed event. The compensation amount is then obtained by:

$$\text{Compensation} = \text{Regional Index} \times \text{Severity Coefficient} \times \text{Unit Amount}$$

We adopted the following structure:

- **No weather alert or minor event**: coefficient = 0 (no compensation)
- **Yellow alert (low risk)**: coefficient = 0.5
- **Orange alert (moderate risk)**: coefficient = 1
- **Red alert (extreme event or severe impact)**: coefficient = 1.5
- **Exceptional event (major natural disaster)**: coefficient = 2

This method makes it possible to automatically adjust the level of compensation according to the severity of the observed situation, while relying on an objective, regionalized index. It also ensures fair, transparent, and understandable pricing for policyholders, and improves risk pooling at the territorial scale.

3.3.3 Application to a case study

On Saturday, April 19, 2025, the Occitanie region was largely placed under yellow alert by Météo France due to adverse weather conditions. According to the bulletin published that day, all departments in the region, except for Gers, were under at least one alert, particularly for rain and flooding.

Step 1: Calculation of the adjusted regional index:
We considered the three main categories of climate risks: physical, liability, and transition. These categories do not all carry the same weight in such an event, and a differentiated weighting is needed to reflect their actual impact.

Physical risk, which includes the direct effects of a climate event (such as floods or heavy rainfall), is the most relevant for a flooding event. It was thus weighted at 70%. Liability risk, though secondary, remains significant: damages to third parties or resulting legal claims justify a weighting of 20%. Finally, transition risk, referring to socio-economic and legal implications, is less directly involved in such a punctual situation and was therefore limited to 10%.

We apply these weights to the scores retrieved in the previous section.

Physical	**Liability**	**Transition**
0.318662	0.523109	0.557265

Thus, the computed score is:

$$\text{Index}_{\text{Occitanie, Flooding}} = 0.70 \times 0.318662 + 0.20 \times 0.523109 + 0.10 \times 0.557265 = 0.3834117$$

Step 2: Calculation of the compensation
The entire region was placed under yellow alert; the corresponding severity coefficient is 0.5.
Based on a study of the average cost of flood-related claims in the Gard department, we estimate that each flood-related claim costs 7,850 €. Thus:

$$\begin{aligned}\text{Compensation}_{\text{Occitanie, Flooding}} &= \text{Index}_{\text{Occitanie, Flooding}} \times \text{Coefficient}_i \times \text{Unit Amount} \\ &= 0.3834117 \times 0.5 \times 7,850 \\ &= 1,504.90\end{aligned}$$

Therefore, for each declared and validated claim in Occitanie, a compensation of 1,504.90 € would be paid.

Step 3: Calculation of the pure premium
As of now, we do not have comprehensive data to accurately estimate the probability that a yellow alert for rain-flooding will be triggered in Occitanie.
However, it is reasonable to assume that insurance companies using this pricing approach would rely on detailed historical archives to robustly determine this probability.
We made an assumption based on identified occurrences during the year 2024. According to available bulletins, four yellow flood alerts were reported. We thus estimate an approximate annual probability of such an alert as: $\mathbb{P}(\text{Alert}_i) = \frac{4}{365} \approx 0.011$.

Applying this estimate to our model, we obtain:

$$\begin{aligned}\text{Pure Premium} &= \mathbb{P}(\text{Yellow Alert}) \times \text{Compensation} \\ &= 0.011 \times 1504.90 \\ &= 16.55\end{aligned}$$

Therefore, the client would need to pay a pure premium of 16.55 € to be insured against flood risks under yellow alert conditions.

Discussion and Perspectives for Improvement

This project was based on the construction of a climate risk score and its integration into an insurance pricing framework. While our approach provides a solid foundation, several limitations must be acknowledged and open avenues for improvement.

First, our data sources were sometimes incomplete or aggregated at a level that limited the precision of the analyses. In particular, the **liability score** proved difficult to assess, as it relies on rare or non-standardized data, such as the frequency of climate-related litigation at a regional scale.

Moreover, in the computation of the pure premium, the risk weightings we used (70% physical, 20% liability, and 10% transition) were justified qualitatively, based on our proper reasoning. However, these proportions could be optimized further using real impact analyses or statistical learning methods based on historical claim data.

The probability of climate alerts (such as yellow flood warnings) was estimated based on a very limited sample and only one year of observation. A more robust model would require historical alert data over several years to improve reliability.
Another assumption that could be refined is the **linearity** between the risk score and the compensation amount. In practice, the relationship between climate impact and financial loss is often non-linear, and adopting a non-linear payoff function might better reflect the true economic consequences of extreme events.

Finally, for our method to be operationally implemented by an insurance company, it would require access to reliable, high-frequency, and regionally detailed climate data. Additionally, regular updates of the score would be necessary to keep up with evolving climate conditions.

Future perspectives could include the integration of a temporal dimension to capture climate trends over time, making the score dynamic and responsive to long-term changes.

Conclusion

In this research project, we have developed a regional climate risk score through a combination of statistical and actuarial methods. This score was designed to synthesize multiple climate-related, socio-economic, and environmental variables, with the goal of providing insurers with a meaningful tool to better assess territorial vulnerability.

We have proposed a pricing method based on this score, inspired by the principles of index-based insurance. The proposed methodology allows for transparent, regionally adapted, and scalable climate risk pricing, with potential for real-world applications in insurance products.

An application to a concrete case in the Occitanie region illustrated the practical feasibility of our score and pricing method. While the results are promising, we have also highlighted limitations regarding data quality, risk weightings, and modeling assumptions. These discussions open up several paths for improvement, particularly in making the score dynamic over time and improving the robustness of the actuarial assumptions.

This project allowed us to mobilize key actuarial skills while addressing a major societal issue: adapting to climate change. We believe that such tools, once refined and enhanced with more comprehensive data, could become an essential part of tomorrow's climate insurance strategies.

Beyond the technical dimension, our project also highlights the growing role actuaries are called to play in addressing climate challenges. By integrating climate risk into pricing tools, insurers can contribute to building more resilient and equitable societies. Further development of our score, including temporal dynamics and more granular data, could pave the way for operational applications in real-world insurance products.

We would also like to express our sincere thanks to our supervisor, Professor Anne Eyraud, for her valuable guidance throughout this project, particularly in the development of the pricing methodology. Her insights helped us better structure our actuarial reasoning while keeping our model both rigorous and accessible.

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appendix A

Maps and Graphs

Contents

A.1 Descriptive Data Analysis

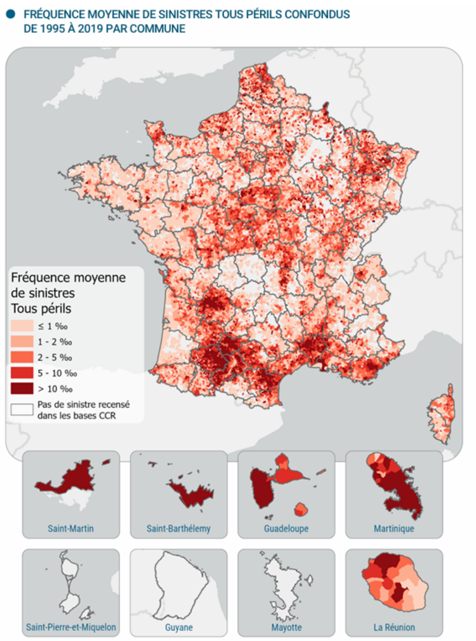

Figure A.1: Map of the frequency of natural disasters

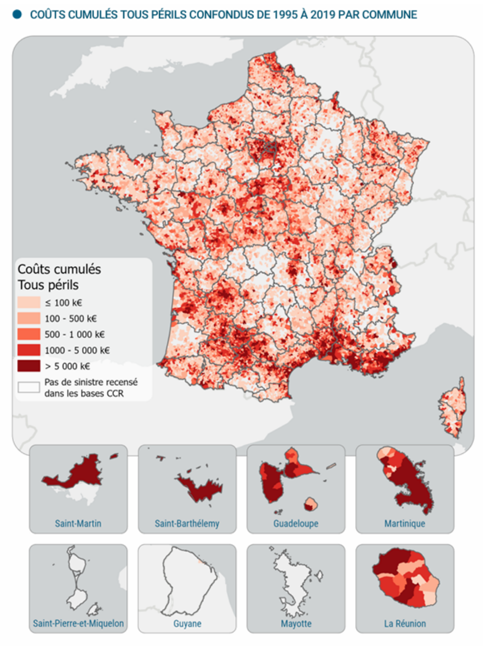

Figure A.2: Map of the distribution of climate risk costs

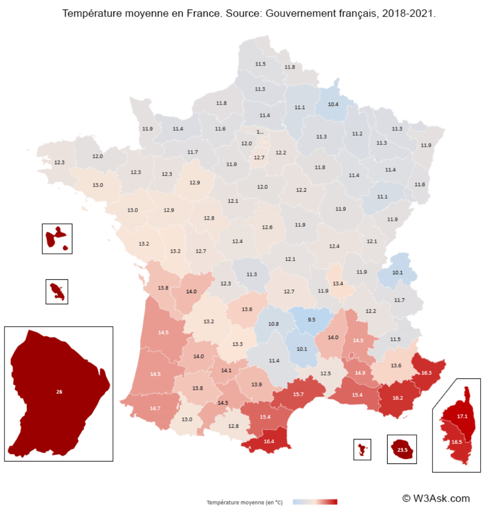

Figure A.3: Map of average temperature in mainland France

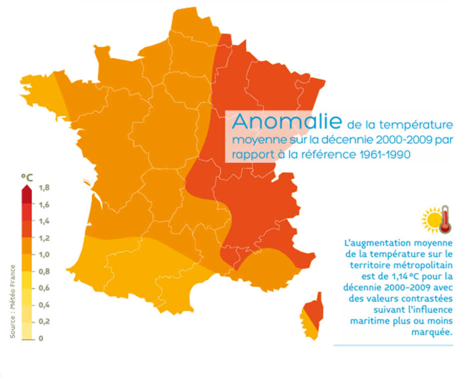

Figure A.4: Map of temperature anomalies in France

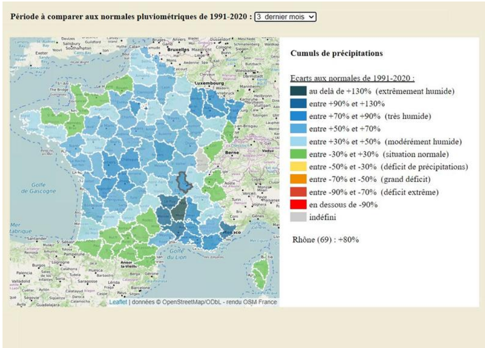

Figure A.5: National map of precipitation levels

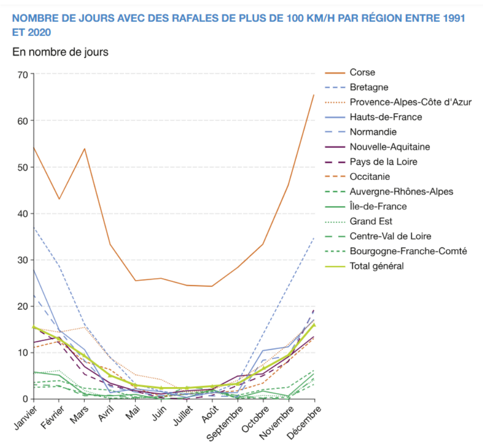

Figure A.6: Number of days with wind gusts over 100 km/h by region (1991–2020)

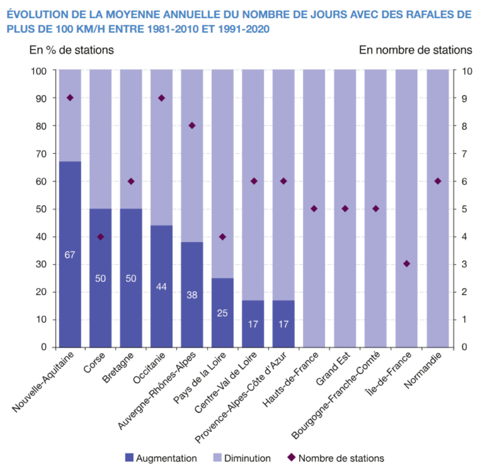

Figure A.7: Change in annual average number of days with wind gusts over 100 km/h (1981–2010 vs. 1991–2020)

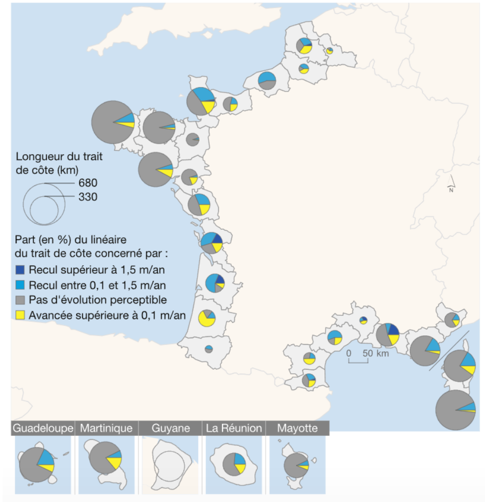

Figure A.8: National map of shoreline evolution by coastal departments

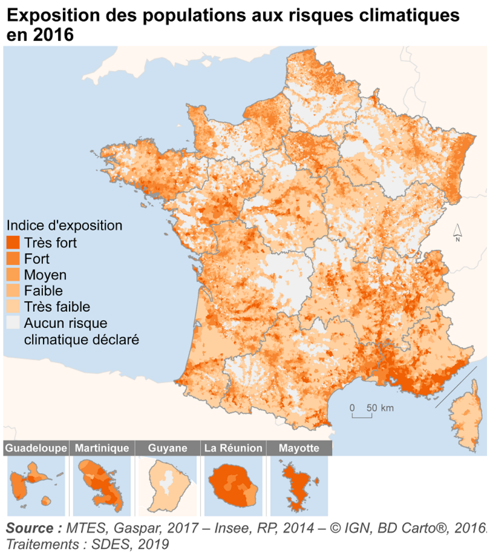

Figure A.9: Population exposure to climate risks

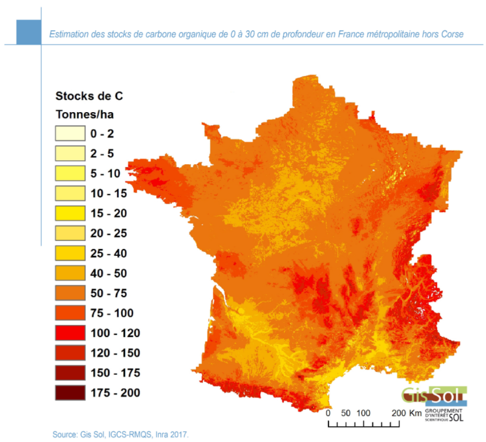

Figure A.10: Map of estimated organic carbon stocks in mainland France

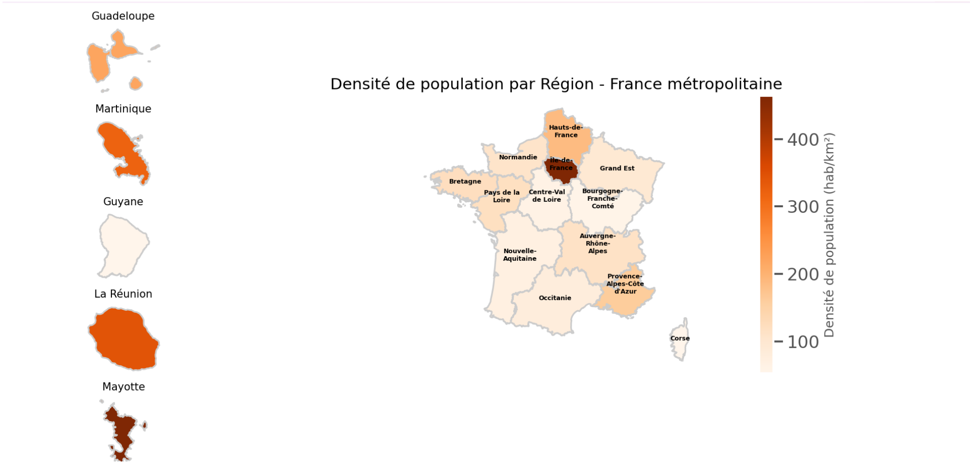

Figure A.11: Map of population density in France

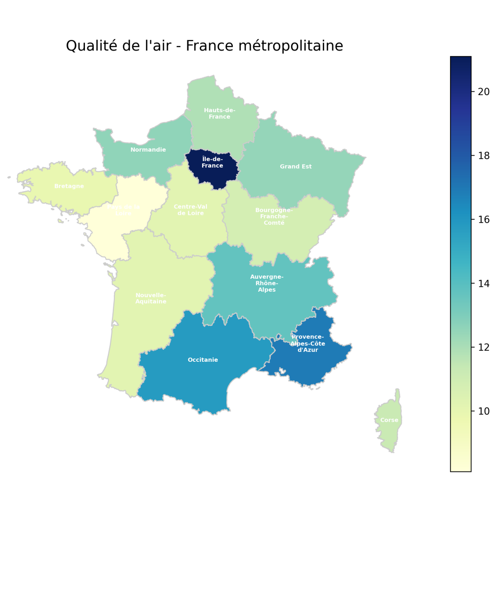

Figure A.12: Map of air quality

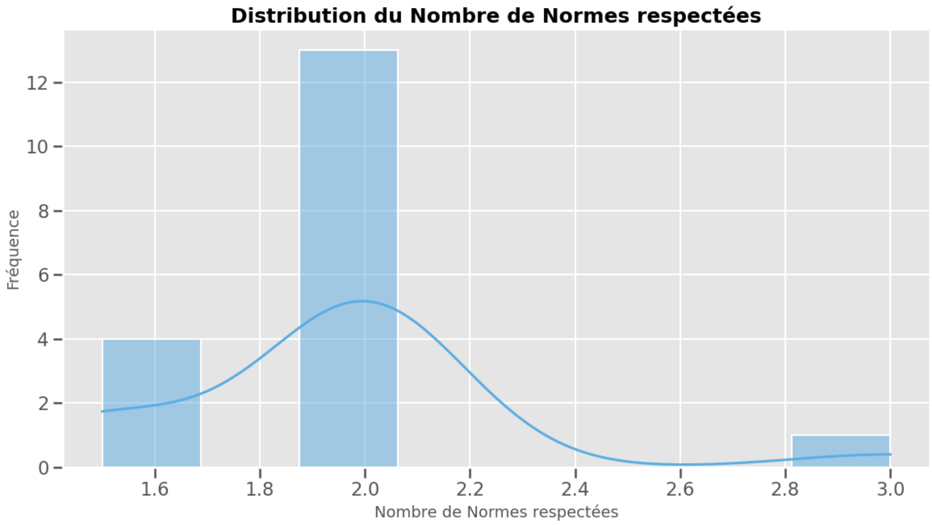

Figure A.13: Histogram of compliance with construction standards

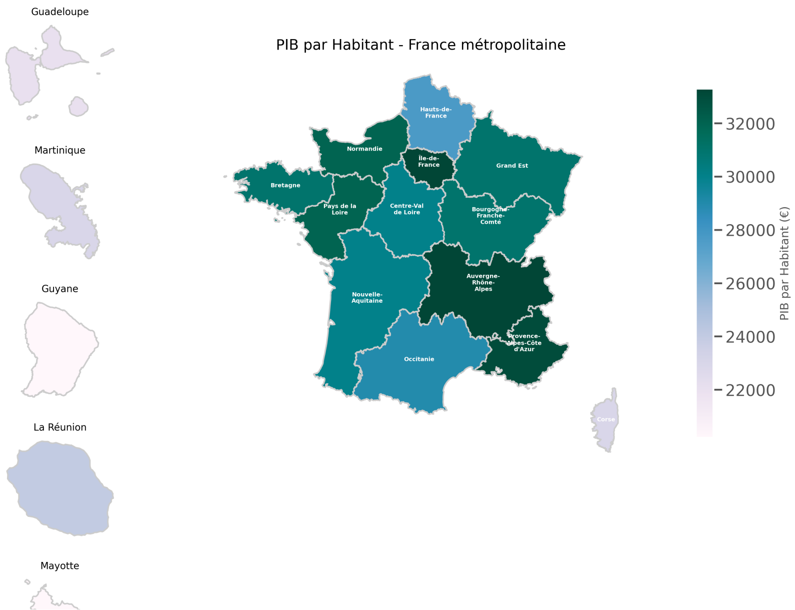

Figure A.14: Map of GDP per capita

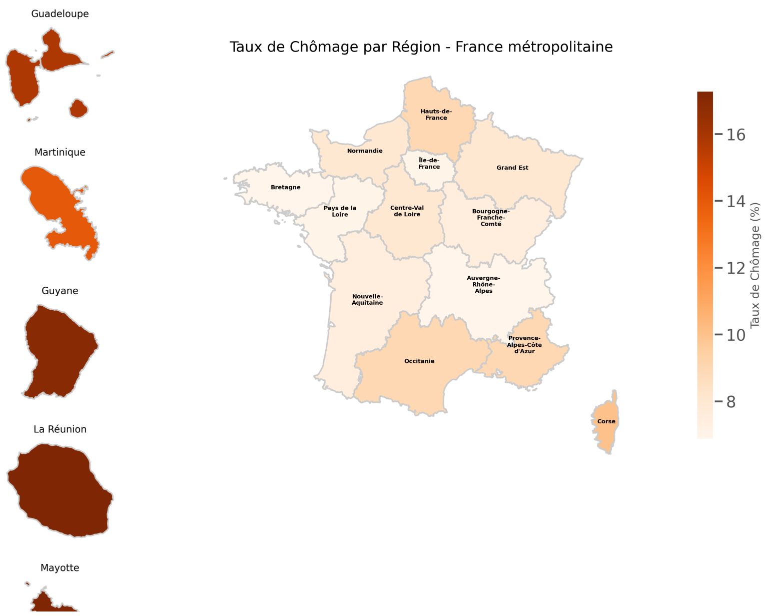

Figure A.15: Map of unemployment rate

A.2 Multivariate Analysis and Clustering Methods

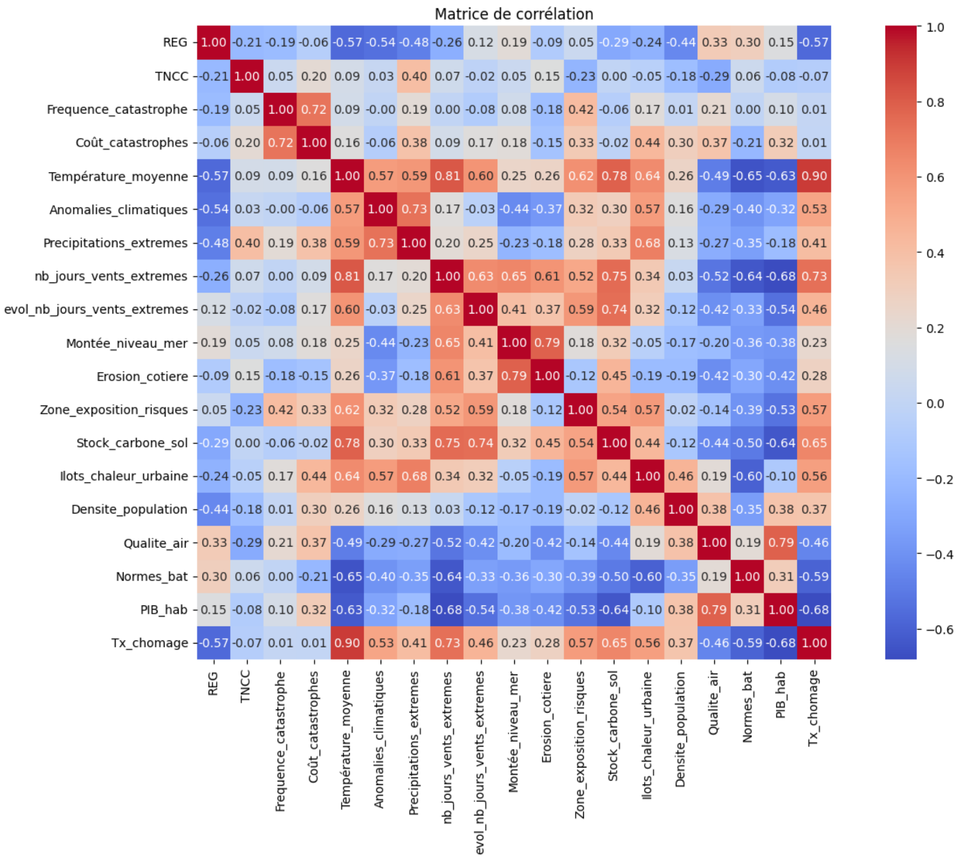

Figure A.16: Correlation matrix of the dataset variables

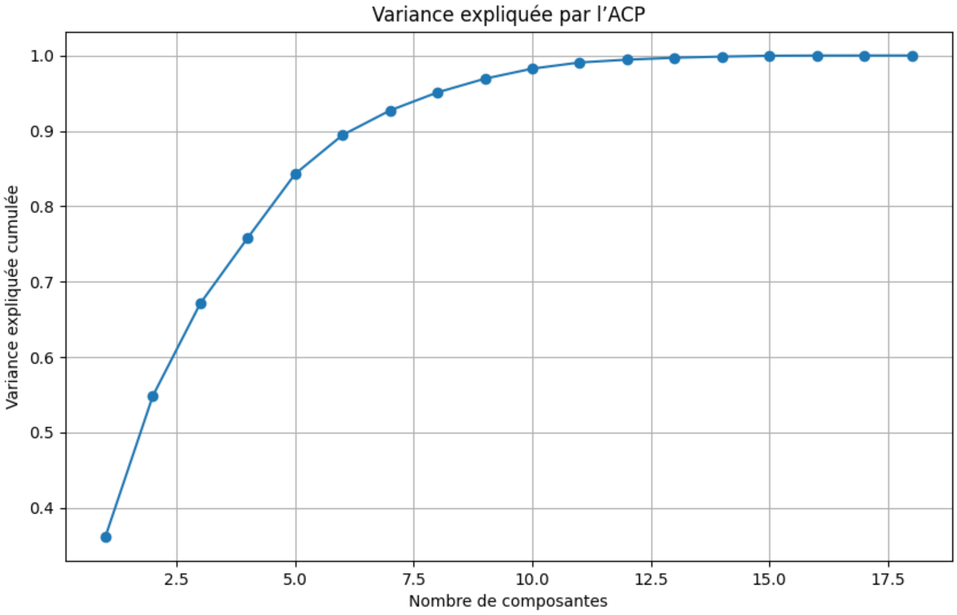

Figure A.17: Principal Component Analysis (PCA) of the dataset

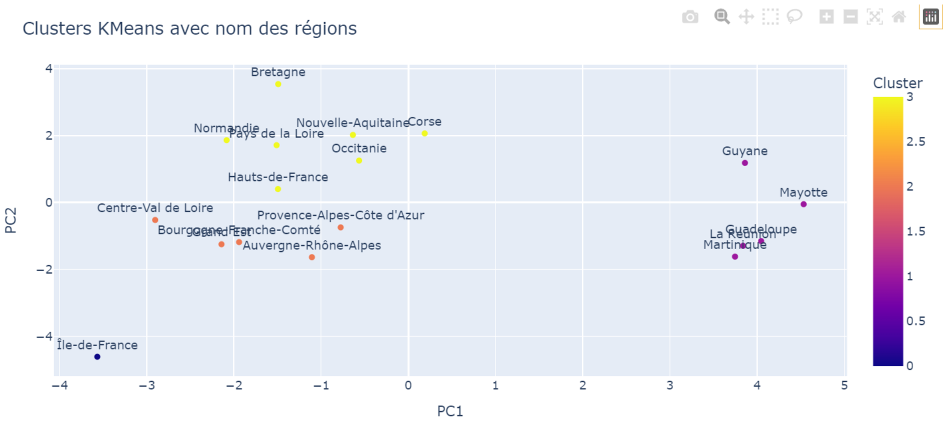

Figure A.18: k-means clustering of the regions in the dataset

appendix B

Project organization and discussions with our academic supervisor

Contents

This part of the document describes how we worked together to carry out our TER on the development of a climate risk index for insurance pricing. Our group consists of Line Bransolle from M1 ES, Ambrune Debroux, and Coralie Rames from M1 Actuariat. Throughout our work, we were supervised by Anne Eyraud, professor of stochastic processes and option theory for the M1 Actuarial Science class.

We decided to work together for several reasons. First, we were all interested in exploring climate-related topics in insurance. Secondly, our friendship was an asset that allowed us to collaborate effectively and support each other in our tasks. Finally, we are all interested in and aware of climate change, making this an obvious choice for us to explore in order to gain a new perspective in our studies.

B.1 Resources Used

To organize our work clearly and ensure we wouldn't duplicate efforts or waste time on the same tasks, we used a tool called *Trello*. It allowed us to create to-do lists and divide tasks among the three of us. This way, when working individually, we could see what each team member had already completed and avoid repeating tasks unnecessarily.

In parallel, we shared a *Google Drive* folder where we stored various documents. For example, we shared studies and scientific articles that we found relevant to our work, drafts of each part of our project, and the evolving outline of the TER.

Since the goal of the TER was to create a climate risk index for insurance pricing, we needed to find a database containing as much relevant information as possible for the theoretical part of our work. For this, we consulted multiple websites to find suitable data sources. We can mention *data.gouv* or *Copernicus*, but in the end, we used any reliable sources like *data.gouv*,obviously, but also *developpement-durable.gouv.fr* for example.

In order to make the map modeling effective in displaying the results we intended to highlight, and to carry out the score calculations, we used VS Code and developed the entire code in Python.

Finally, because our work had to be written in *LaTeX*, we collaborated on a shared document via *Overleaf* to draft and finalize everything.

B.2 Timeline of Tasks

The first step in our reflection for the TER was to choose the main topic that we wanted to explore in more depth. As Miss Anne Eyraud had suggested working on a subject related to climate change and insurance, it quickly became obvious to us that we wanted to collaborate with her on this project.

We then had to define more concretely what we would actually do. During her bachelor's degree, Line worked on a project to create a website, along with friends, featuring an interactive map designed to help users plan environmentally responsible vacations. They developed an environmental score influenced by several criteria and assigned a grade to each country to indicate its environmental responsibility.
We thought this was a great idea and decided to adapt it to the context of insurance for our TER.

Once the subject was chosen, we had to find out whether an existing database could help us build our own risk score. However, this turned out to be more difficult than expected, and we realized we would need to construct our own database from scratch. This became a major part of our work. We had to select the relevant criteria to include, and then search for available information for each of them in order to build a scale and create a usable database. To be efficient, we split the research work and managed to complete the database as early as possible so we could focus on developing a pricing system adapted to insurance companies.

Once the database was created, we proceeded to analyze it and began developing the climate score. Through our research on existing scoring methods, we identified three approaches that could be adapted to our work. We then implemented these three different scores using Python.

Finally, we aimed to apply the constructed scores to a concrete example of a pricing method to assess their practical relevance. We explored several pricing strategies and selected one that was consistent with our index structure. Once the scores had been computed, we chose a representative French region whose scores varied significantly across categories, in order to illustrate how the premium would be calculated in practice. This led us to the pricing example for the Occitanie region.

To conclude this research project, we briefly discuss potential improvements to our approach, and reflect on what would be needed to enhance the applicability of this methodology in a real-world insurance context.

B.3 Interaction with the Supervising Professor

We were mostly able to work independently throughout this project and found most of the information we needed online. However, whenever we required a more academic perspective or couldn't find the answers we were looking for, we knew that our supervisor was available to guide us and suggest relevant readings.

Her help was particularly valuable when we reached the pricing section of our work. She provided us with the necessary resources and explanations to get started in the right direction.